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Why global temperature has risen since 1976 — and what made the 1940s bump

A detailed look at each driver behind the curve on the Planet tab: how much it contributed, when, and how sure scientists are.

The general statement. Since 1976 the world has warmed by about 1.0 °C, at roughly five times the rate of 1880–1975. The long-term rise is caused by greenhouse gases from burning fossil fuels; that is settled. The *timing* of the turn in the mid-1970s came from two other things happening at once: air-pollution haze, which had been masking greenhouse warming since the 1940s, stopped growing in Europe and North America; and the Pacific flipped into a warm phase in 1976–77. On top of the steady climb, El Niño years make spikes, and big volcanic eruptions make dips. The Sun did not drive it. Since about 2010 warming has run faster again. Sharp cuts in sulphur pollution from China and from shipping removed more of the mask, and the planet now reflects less sunlight because some low clouds have thinned; how big the speed-up is, and how much each cause contributed, are still debated.

Who did what is broken down in section 9: each activity, gas and country, the events behind them, and what would reduce each part.

1. What the record shows

NASA GISTEMP v4, annual means, anomalies against 1951–1980 (the same data as the Planet chart):

Period	Linear trend	Note
1880–1975	+0.04 °C per decade	Includes the 1940s bump and the flat 1950s–70s
1976–2025	+0.21 °C per decade	About +1.0 °C over the period
1996–2025	+0.25 °C per decade	The most recent 30 years

Period	Linear trend	Note
Decade means	1971–80: +0.06 °C · 2016–25: +1.01 °C	A rise of about 0.96 °C

Global surface temperature, 1880 to the last complete year

Annual means with straight-line trends before and after 1976. Hover for values.

Source: NASA GISTEMP v4 (complete calendar years only), anomalies against 1951–1980. Trend lines are least-squares fits computed by this site.

Measuring from 1976 alone overstates the rise: 1976 (−0.10 °C) came at the end of a long 1973–76 La Niña, a cold phase of the Pacific. On the IPCC's 1850–1900 baseline, the latest international assessment puts the 2015–2024 decade at 1.24 °C above pre-industrial and 2024 alone at about 1.52 °C, of which an estimated 1.36 °C was human-induced.[1]

2. Greenhouse gases — the main driver

- **CO_2 in the air:** 332 ppm in 1976 and 427 ppm in 2025 at Mauna Loa (annual means, NOAA). That is a 29% rise in 49 years.
- **Emissions behind it:** fossil CO_2 emissions more than doubled, from 17.9 billion tonnes in 1976 to 38.6 billion tonnes in 2024. Coal rose from 6.2 to 15.8 billion tonnes, a record, oil from 8.5 to 12.5 and gas from 2.4 to 8.0. Deforestation and other land-use change added another 4.6 billion tonnes in 2024 (Global Carbon Budget 2025).[20]
- **Total heat-trapping effect:** NOAA's Annual Greenhouse Gas Index rose 54% between 1990 and 2024. It took about 240 years to reach the 1990 level and only 34 more years to add another 54%. CO_2 contributes most, then methane.[3]
- **CFCs — an extra push in the 1970s and 1980s:** the ozone-destroying gases are also strong greenhouse gases. Their warming effect grew fast from 1950 to about 1990, then the Montreal Protocol (1987) stopped the growth; CFC-11 has fallen since the

mid-1990s.[4] So these gases added to the steep climb of 1975–1990, and their contribution has levelled off since.

- **How much warming:** the IPCC assesses that well-mixed greenhouse gases likely caused 1.0–2.0 °C of warming from 1850–1900 to 2010–2019 — more than the total observed, because aerosols cancelled part of it.[5]

Where the CO_2 came from, 1950–2024

Annual CO_2 emissions by fuel, plus deforestation and other land-use change, stacked.

Source: Global Carbon Budget 2025, via Our World in Data (World totals). Land-use change emissions carry an uncertainty of roughly ±50%.

The heat-trapping push from each greenhouse gas, 1900–2024

Effective radiative forcing relative to 1750, in watts per square metre of Earth's surface.

Source: Indicators of Global Climate Change 2024 (Forster et al. 2025), best estimates. Ozone-depleting gases more than doubled their push between 1976 (0.15 W/m²) and 1990 (0.32 W/m²), levelled off around 2000–2010 and were 0.34 W/m² in 2024; their HFC replacements reached 0.05 W/m².

Fingerprint: the stratosphere has cooled while the surface warmed. A brighter Sun would warm both; heat-trapping gases near the surface produce exactly this split.[6]

3. Aerosols — the mask that slipped

Sulphur dioxide from coal and oil forms tiny sulphate particles that reflect sunlight and brighten clouds. They cool the planet, and they fall out of the air within days, so their effect tracks current emissions.

- **1940s–1970s:** rapidly rising pollution after the war largely offset greenhouse warming — the flat middle of the chart.
- **The turn:** sulphur emissions peaked around 1970 in North America and around 1980 in Europe, after clean-air laws; the global total peaked in the 1980s.[7] Once the mask stopped thickening, greenhouse warming showed through.
- **China:** emissions peaked in 2006; East Asia then cut sulphur dioxide by about 75% in roughly 15 years. A 2025 study (Samset and co-authors, *Communications Earth & Environment*) concludes this cleanup has likely driven much of the recent speed-up in warming.[8]
- **Shipping, 2020:** the International Maritime Organization cut the sulphur allowed in ship fuel. A 2024 study led by Pacific Northwest National Laboratory found the expected loss of cooling over busy shipping lanes matches observed cloud changes and contributed to 2022–2023 Northern Hemisphere heat, but is too small to be the only cause; the authors call their estimate possibly conservative.[9]
- **In the emissions data:** global sulphur dioxide peaked in 1979 at 138 million tonnes and fell to 69 million tonnes by 2023. Power plants alone peaked later, in 1990, at 59 million tonnes. Ship emissions fell from 10.4 million tonnes in 2019 to 2.0 million tonnes in 2020, an 81% cut in one year (CEDS).[22]
- **How much cooling in total:** the IPCC puts the cooling from aerosols and other non-greenhouse human factors at 0.0–0.8 °C by 2010–2019.[5] That wide range is the largest single uncertainty in explaining the past century — and the reason the size of the recent acceleration is disputed.

Sulphur dioxide emissions by source, 1950–2023

The pollution behind the aerosol “mask”, stacked by sector. Hover for values.

Source: Community Emissions Data System (CEDS) v2025-03-18, global totals by sector, as used by the Indicators of Global Climate Change. Regional detail (China, Europe, North America) is not in this file; see sources 7–8.

4. The Pacific's 1976–77 shift — the timing

In 1976–77 the North Pacific changed regime: the Pacific Decadal Oscillation switched from its cool to its warm phase, and the Aleutian Low strengthened for about three decades (1977–2005), after a weak period from 1947 to 1976. It altered how El Niño evolves and how it affects weather across the Americas, Asia and Australia.[10] Earlier shifts of the same kind happened in 1924–25 and 1946–47.

The PDO does not add heat to the planet; it moves it between the ocean and the air. So it explains *why the climb started so sharply around 1976–77*, not why it kept going for fifty years. The IPCC notes the PDO appears to respond to the same forcing as these shifts rather than trigger them.[10]

5. Natural ups and downs on top of the trend

El Niño and La Niña

Strong El Niños lift global temperature for about a year, usually peaking a few months after the Pacific does. The big ones in the record since 1976, by peak ONI (ERSSTv6, 3-month means): 1982–83 +2.14, 1997–98 +2.37, 2015–16 +2.59, 2023–24 +1.99 — and now 2026, already +2.16 for July–September. Each lifted global temperature in the year that followed (1983 less than expected, because El Chichón's haze offset it); La Niña years give the dips.

Volcanoes

- **El Chichón, Mexico, 1982** and **Pinatubo, Philippines, 1991** put sulphur high into the stratosphere. Pinatubo injected about 15 million tonnes of sulphur dioxide and cooled the globe by about 0.6 °C for around 15 months; the effect lasted almost two years.[11] These eruptions explain the dips of 1982–84 and 1992–93.
- **Hunga Tonga, 2022** added a huge amount of water vapour, a greenhouse gas, as well as sulphate. A NASA-led study found its net effect was a slight *cooling*, so it did not cause the 2023–24 heat.[12]

The Sun

Solar output follows its 11-year cycle with no net increase since the 1950s, and satellites since 1978 show no upward trend.[6] The recent solar maximum (cycle 25) adds a small, temporary push; it cannot explain a 50-year rise.

Natural versus human push on the climate, 1950–2024

Effective radiative forcing relative to 1750. Volcanic eruptions (Agung, Bali, 1963; El Chichón, Mexico, 1982; Pinatubo, Philippines, 1991) make short negative spikes; the Sun moves in an 11-year cycle; the human push climbs.

Source: Indicators of Global Climate Change 2024 (Forster et al. 2025). The volcanic series is positive in quiet years because it is measured against the 1750 baseline, which included background eruptions.

6. The 1998–2012 slowdown

Surface warming slowed for about a decade after the 1998 El Niño. The heat did not disappear: Pacific trade winds strengthened from the 1990s and pushed extra heat into the subsurface western Pacific (England and co-authors, *Nature Climate Change*, 2014). The authors expected it to return once the winds eased.[13] It did: 2014–2016 brought three consecutive records.

7. 2023–2026: a jump, and an acceleration debate

- **The planet is absorbing more energy:** satellite and ocean data show Earth's energy imbalance roughly doubled between 2005 and 2019 — from fewer reflective clouds and less sea ice, more greenhouse gases and water vapour, and the PDO's swing to a warm phase (Loeb and co-authors, *Geophysical Research Letters*, 2021).[14]
- **The 2023 gap:** greenhouse gases, El Niño and volcanoes together leave about 0.2 °C of the 2023 record unexplained. A *Science* study led by the Alfred Wegener Institute attributes most of it to a decline in certain low clouds, which made the planet less reflective.[15] Whether that cloud loss is a response to cleaner air, natural variability or a warming feedback is still open.

- **How fast now?** The international indicators team puts the human-induced rate at 0.27 °C per decade for 2015–2024, unprecedented in the instrumental record, partly due to weaker aerosol cooling.[1] Foster and Rahmstorf (*Geophysical Research Letters*, 2026) remove El Niño, volcanic and solar effects and find about 0.2 °C per decade for 1970–2015 and about 0.35 °C per decade since 2015.[16] Others disagree on the size: Michael Mann sees no acceleration beyond the aerosol effect; Zeke Hausfather prefers about 0.27 °C; Piers Forster prefers "unprecedented rates" to "acceleration".[16] Over 1970–2008 the rate was about 0.18 °C per decade, so warming is now faster on every estimate; the debate is about by how much.[17]

8. The warm bump around the Second World War

The early-1940s spike on the chart is partly real and partly a measurement artefact:

- **Real:** a strong, long El Niño ran from 1940 to 1942. Brönnimann and co-authors (*Nature*, 2004) reconstructed its global effects, including a cold Europe and a warm Alaska, and described it as a recurring extreme state of the global climate system linked to strong El Niños.[18]
- **Artefact:** most ocean temperatures came from ships. Before the war many fleets measured sea water in canvas buckets, which cool by evaporation and read low — about 0.4 °C cooler on average than modern methods. From about 1939–41 the record came almost entirely from US ships, which read engine-intake water and run warm. After 1945 British, Dutch and Japanese bucket readings returned, producing an artificial drop. Thompson and co-authors (*Nature*, 2008) identified the 1945 discontinuity. Modern ocean datasets, including the ERSST data inside GISTEMP, now apply bias corrections for these changes in method, but the wartime years remain among the least certain in the record.[19]

So the 1940s peak is not evidence of a warming episode like today's; it is an El Niño seen partly through biased instruments. The later 1950s–70s plateau is real and reflects rising aerosol pollution.

9. Who did what

A statement such as "people warmed the planet by 1 °C" is true but tells you nothing you can act on. This section works like a household energy log: what happened, who did it, how much each part added, and what would reduce it. It follows one chain:

1. Activity	2. Emission	3. Push	4. Warming

Burning coal, driving, clearing forest, raising cattle	CO_2, methane, sulphur dioxide, CFCs	Extra energy trapped, in watts per m^2 (radiative forcing)	Temperature rise, in °C, as the oceans catch up

9.1 The overall verdict

Warming from 1850–1900 to 2010–2019 (IPCC AR6, Summary for Policymakers, A.1.3):[5]

Factor	Contribution	
Observed warming (2011–2020)	+1.09 °C (0.95 to 1.20)	
All human influence	+1.07 °C best estimate (0.8 to 1.3)	settled
Well-mixed greenhouse gases	+1.0 to +2.0 °C	settled
Other human drivers, mainly aerosols	−0.8 to 0.0 °C	wide range
Natural drivers (Sun, volcanoes)	−0.1 to +0.1 °C	settled
Internal variability (El Niño, PDO ...)	−0.2 to +0.2 °C	settled

9.2 What pushed the climate since the 1970s

Between the 1971–80 and 2015–24 decade averages, the human push on the climate grew by **2.20 W/m²**. Spread over the whole Earth, that is about 1,100 trillion watts of extra energy kept in the system, continuously.[21]

- **Carbon dioxide: +1.20 W/m², 55% of the increase.** The single biggest item.
- **Less aerosol haze: +0.27 W/m², 12%.** Not a new emission but a lost shield: cleaner air let more sunlight through (section 3).
- **CFCs, HCFCs and HFCs: +0.26 W/m², 12%.** Refrigerants, aerosol-can propellants and foams. Most of it came before 1990; the Montreal Protocol then stopped the growth of the ozone-depleting kinds.
- **Methane: +0.15 W/m², 7%**, and **low-level ozone (smog): +0.14 W/m², 7%**. Ozone forms in the lower atmosphere from methane, traffic and power-plant nitrogen oxides, and carbon monoxide.
- **Nitrous oxide: +0.11 W/m², 5%**, mainly from farming. **Aircraft contrails: +0.04 W/m², 2%.**
- **Natural factors were slightly negative:** the Sun −0.03 and volcanoes −0.02 W/m² between the same decades.

Change in the push on the climate, 1971–80 to 2015–24

Decade-average effective radiative forcing; red pushes towards warming, blue towards cooling. Hover a bar for its share.

Source: Indicators of Global Climate Change 2024 (Forster et al. 2025), best-estimate forcing by agent. Decade averages are used because single years mislead for the Sun (11-year cycle) and volcanoes.

9.3 How the push became warming

Jones and co-authors convert each country's emissions of the three main greenhouse gases into warming, year by year. On their numbers, these three gases had caused **0.75 °C** of warming by 1976 and **1.68 °C** by 2024: **+0.93 °C** in 48 years. Carbon dioxide added 0.71 °C of that, methane 0.17 °C and nitrous oxide 0.05 °C.[27]

This split covers only those three gases; it leaves out the CFC, ozone and aerosol effects in 9.2, which have no country breakdown. It is the standard published way to answer "whose emissions caused how much".

Warming caused by CO_2, methane and nitrous oxide, 1851–2024

Stacked, in °C relative to 1850. Hover for values.

Source: Jones et al., National contributions to climate change (2025 data release), via Our World in Data.

9.4 Which activities

Carbon dioxide, 1976–2024: 1,636 billion tonnes in total. Coal 525 billion tonnes (32%), oil 500 (31%), deforestation and other land-use change 295 (18%), natural gas 242

(15%), cement 44 (3%), gas flaring 16 (1%) and other industry 14 (1%).[20]

CO_2 emitted from each source, 1976–2024

Cumulative billion tonnes. Cumulative emissions, not annual ones, set the warming from CO_2.

Source: Global Carbon Budget 2025, via Our World in Data.

Methane from human sources rose from 260 to 347 million tonnes a year between 1976 and 2023 (+33%).[22]

- **Livestock** (digestion and manure): 88 → 116 million tonnes, the largest source.
- **Landfills and wastewater:** 35 → 65 million tonnes, nearly double.
- **Coal mining:** 25 → 52 million tonnes, doubled with coal output.
- **Oil and gas:** 82 → 75 million tonnes in this inventory. The IEA's satellite-informed estimate is higher, 81 million tonnes from oil and gas in 2025 alone, so treat this line as a lower bound.[24]
- **Rice paddies:** 22 → 25 million tonnes.

Natural sources such as wetlands are not included; they are not human activities, although warming may be increasing them.

Methane emissions from human activities, 1970–2023

Million tonnes per year, stacked by source.

Source: CEDS v2025-03-18 global emissions by sector. Livestock = enteric fermentation plus manure management.

9.5 Which countries

Of the 0.93 °C added by the three gases since 1976:[27]

- **China: 0.171 °C (18%).** Fossil CO_2 nearly tripled in 13 years, from 3.6 billion tonnes in 2000 to 9.9 billion in 2013, mostly coal.
- **United States: 0.126 °C (14%).** Emissions per person are three times the world average: 14.2 tonnes of CO_2 in 2024, against 4.7.
- **Brazil: 0.060 °C (6.5%).** Mostly deforestation: 77 of its 94 billion tonnes of CO_2 since 1976 came from land-use change.
- **Russia: 0.052 °C (5.6%).**
- **India: 0.050 °C (5.4%).** Fossil CO_2 grew 13-fold, from 0.24 billion tonnes in 1976 to 3.2 billion in 2024, but is still 2.2 tonnes per person.
- **Indonesia: 0.036 °C (3.9%).** Mostly land-use change: 42 of its 58 billion tonnes of CO_2.
- **Places 7–22:** Japan 0.024 °C; Canada 0.018; Germany 0.017; Mexico 0.014; Australia, Saudi Arabia, South Africa and Iran 0.012 each; South Korea and Ukraine 0.010; Italy, Nigeria, Pakistan, Thailand, Turkey and the United Kingdom 0.009 each.
- **The 193 other countries together: 0.238 °C (26%).**

By income group: high-income countries (1.4 billion people) added 0.36 °C and upper-middle-income countries, including China (2.9 billion), 0.39 °C. Lower-middle-income countries (3.1 billion) added 0.13 °C and low-income countries (0.8 billion) 0.04 °C.[27]

Warming added 1976–2024: the top 20 countries

°C from CO_2, methane and nitrous oxide emitted on each country's territory; labels show the share of the 0.93 °C total. Six countries tie at 0.009 °C for 17th place, so 22 are shown. Hover a bar for population and emissions per person.

Source: Jones et al. (2025 release) and Global Carbon Budget 2025, via Our World in Data. Counted where the emissions happened; goods made in one country and used in another stay with the producer. CO_2 figures include land-use change.

9.6 The underlying events

When	What happened	Measured change
1976–2024	World fossil fuel use grows	Fossil CO_2 17.9 → 38.6 billion t/yr; coal at a record 15.8 billion t in 2024[20]
1979	Europe's emissions peak; global sulphur pollution peaks	EU-27 fossil CO_2 peaks at 4.1 billion t (2.4 in 2024, −41%); world SO_2 peaks at 138 million t[20,22]
1976–1990	CFC use in fridges, sprays and foams	Push from ozone-depleting gases doubles, 0.15 → 0.32 W/m²; Montreal Protocol signed 1987[21]
1997	Indonesia's land-use emissions peak	2.4 billion t CO_2 (0.6 in 2024)[20]
2000–2013	China's coal-powered industrial boom	Fossil CO_2 3.6 → 9.9 billion t; coal 2.6 → 7.5 billion t[20]
2003	Brazil's land-use emissions peak	Brazil land-use CO_2 3.0 billion t (1.6 in 2024)[20]
2005–2024	US shifts from coal to gas	US coal CO_2 2.14 → 0.73 billion t (−66%); gas 1.16 → 1.75; US total 6.1 → 4.9 billion t[20]
2006 onward	China cuts its sulphur pollution	East Asian SO_2 down about 75% in 15 years, removing part of the cooling mask[8]
1976–2023	More landfills, sewage, coal mines and cattle	Methane: landfills and wastewater 35 → 65 million t; coal mines 25 → 52; livestock 88 → 116[22]
2016	Kigali Amendment to phase down HFCs, the CFC replacements	HFC push 0.05 W/m² in 2024, the fastest-growing halogenated group[21]
2020	Low-sulphur ship fuel rule	Shipping SO_2 10.4 → 2.0 million t in one year (−81%)[22]

9.7 What would reduce each part

Source	Share	What reduces it	Evidence
Coal	32% of CO_2 since 1976; about a third of fossil-fuel methane	Replace coal power with solar, wind and other low-carbon electricity; capture or close off mine methane	IPCC: options costing up to $100 per tonne could halve global emissions by 2030; solar costs fell 85% since 2010[23]
Oil	31% of CO_2 since 1976	Electric vehicles, efficiency, public transport, less flying	IPCC: demand-side changes could cut end-use emissions 40–70% by 2050[23]
Deforestation	18% of CO_2 since 1976	Stop clearing forest	Brazil's land-use CO_2 fell from 3.0 to 1.6 billion t after its 2003 peak[20]
Gas, and methane leaks from oil, gas and coal	15% of CO_2; about a third of human methane	Fix leaks, end venting and routine flaring	IEA: about 70% of fossil-fuel methane (~85 million t) can be cut with existing technology; over 35 million t at no net cost[24]
Cattle and other livestock (digestion)	About 32% of human methane, with manure	Feed additive 3-NOP (sold as Bovaer); more digestible feed; healthier, more productive animals so fewer are needed per litre of milk; breeding for lower-methane animals	3-NOP cut methane from digestion by 33% on average across trials and by 21% in a year-long study; it works better at higher doses and less well on high-fibre diets[28,30]
Manure	Part of the livestock share	Digesters or covered lagoons that capture the gas for energy; shorter storage; solid separation and composting	No single global figure: results depend on climate and farm type[25]
Rice paddies	About 8% of human methane	Alternate wetting and drying: let the field drain until the water is 15 cm below the surface, then	30–70% less methane, no loss of yield and about 30% less water (IRRI)[29]

Source	Share	What reduces it	Evidence
		re-flood, checking the level with a perforated tube	
Landfills and wastewater	About 20% of human methane	Collect landfill gas with wells and burn it for energy or flare it; send food waste to composting or digesters; cover and seal landfill cells	UNEP: human methane from all sources together can fall up to 45% this decade, avoiding nearly 0.3 °C of warming by 2045[25]
HFCs	Fastest-growing halogenated gases	Phase-down under the Kigali Amendment	UNEP: up to 0.5 °C avoided by 2100[26]
Sulphur (the mask)	Cleanup added about 12% of the push since the 1970s	Keep cleaning the air for health; cut methane quickly to offset the lost cooling	Methane works within decades (UNEP, above), while CO_2 cuts take longer to show[25]

About 3-NOP (Bovaer). It is a feed additive given to cows, not added to milk. It blocks an enzyme that methane-making microbes in the cow's rumen need. The reductions above apply only to methane from digestion, not to all the emissions of dairy farming (manure, feed crops, energy). Milk: the additive itself was not detected in milk from treated cows; a breakdown product, NOPA, appears at very low levels. The European Food Safety Authority found no safety concern for consumers, and the UK Food Standards Agency says the milk is safe to drink. The fertility, inhalation and skin warnings on the label are for people handling the concentrated additive, not for milk drinkers.[31] In the United States it is not formally "FDA-approved": in May 2024 the FDA described Bovaer 10 as an unapproved animal drug, said it would not enforce the drug requirements under set conditions, and found it "expected to pose low risk to humans or animals" when used as intended.[32]

10. Settled and still debated

Question	Status
Greenhouse gases cause the long-term rise since the 1970s	settled
Aerosol cleanup and the 1976–77 Pacific shift set the timing of the turn	well supported
The Sun is not a driver of the post-1976 rise	settled
The 1940s bump is part El Niño, part measurement bias	well supported
How large the post-2010 acceleration is (0.27 vs 0.35 °C per decade)	debated
How much of it is aerosol cleanup vs cloud feedback vs natural variability	debated

Sources

1. Forster, P. M. et al. (2025). Indicators of Global Climate Change 2024. *Earth System Science Data* 17, 2641. essd.copernicus.org
2. Global Carbon Budget 2024, historical budget table. openclimatedata.net
3. NOAA Global Monitoring Laboratory (2025). NOAA index tracks the continuing increase in heating from greenhouse gases. gml.noaa.gov
4. IPCC AR4 WG1 (2007), section 2.3.4, Montreal Protocol gases. archive.ipcc.ch
5. IPCC AR6 WG1 (2021), Summary for Policymakers, A.1.2–A.1.3. ipcc.ch
6. NASA. Is the Sun causing global warming? science.nasa.gov
7. Our World in Data. Air pollution: does it get worse before it gets better? ourworldindata.org
8. Columbia Climate School (14 July 2025), on Samset, B. H. et al., *Communications Earth & Environment*. news.climate.columbia.edu
9. Plymouth Marine Laboratory (16 Aug 2024), on "Has reducing ship emissions brought forward global warming?", *Geophysical Research Letters*, doi:10.1029/2024GL109077. pml.ac.uk
10. IPCC AR4 WG1 (2007), section 3.6.3, Pacific decadal variability. archive.ipcc.ch
11. NASA Earth Observatory. Global effects of Mount Pinatubo. science.nasa.gov
12. Texas A&M (July 2024), on Schoeberl, M. et al., *Journal of Geophysical Research: Atmospheres*. artsci.tamu.edu
13. UNSW (Feb 2014), on England, M. H. et al., *Nature Climate Change*. unsw.edu.au
14. AGU (2021), on Loeb, N. G. et al., *Geophysical Research Letters*. news.agu.org
15. Alfred Wegener Institute press release, on Goessling, H. F. et al., *Science* (2024). awi.de
16. Carbon Brief (March 2026). Pace of global warming has nearly doubled since 2015, study says (Foster and Rahmstorf, *GRL*, doi:10.1029/2025GL118804). carbonbrief.org
17. Carbon Brief. Factcheck: why the recent acceleration in global warming is what scientists expect. carbonbrief.org
18. Brönnimann, S. et al. (2004). Extreme climate of the global troposphere and stratosphere in 1940–42 related to El Niño. *Nature* 431, 971. abstract

19. RealClimate (June 2008). Of buckets and blogs, on Thompson, D. W. J. et al., *Nature* 453 (2008). realclimate.org
20. Global Carbon Budget 2025 (Friedlingstein et al.), national and global CO_2 by fuel and land-use change, via Our World in Data. github.com/owid/co2-data; globalcarbonbudget.org
21. Indicators of Global Climate Change 2024: effective radiative forcing time series, best estimates by agent, 1750–2024. Smith, C. et al., ClimateIndicator forcing-timeseries, doi:10.5281/zenodo.10706344
22. Community Emissions Data System (CEDS) v2025-03-18, global emissions by sector (Hoesly, R. M. et al. 2018, *Geoscientific Model Development* 11, 369), as distributed with source 21.
23. IPCC AR6 WGIII (2022), Summary for Policymakers. ipcc.ch; summary in Down To Earth, six takeaways.
24. IEA (2026). Global Methane Tracker 2026: key findings. iea.org
25. UNEP and Climate and Clean Air Coalition (2021). Global Methane Assessment, as reported by WMO. wmo.int
26. UNEP Ozone Secretariat. The Kigali Amendment. ozone.unep.org
27. Jones, M. W. et al. (2023). National contributions to climate change due to historical emissions of carbon dioxide, methane, and nitrous oxide since 1850. *Scientific Data* 10, 155, doi:10.1038/s41597-023-02041-1; 2025 data release zenodo.org, via Our World in Data.
28. Kebreab, E. et al. (2023). A meta-analysis of effects of 3-nitrooxypropanol on methane production, yield, and intensity in dairy cattle. *Journal of Dairy Science* 106(2), 927–936. agris.fao.org
29. International Rice Research Institute. Alternate wetting and drying. ghgmitigation.irri.org
30. van Gastelen, S. et al. (2024). Long-term effects of 3-nitrooxypropanol on methane emission and milk production characteristics in Holstein-Friesian dairy cows. *Journal of Dairy Science* 107(8), 5556–5573. agris.fao.org
31. Reuters Fact Check (17 Jan 2025). Bovaer's active ingredient not detected in dairy products, as republished by The Cattle Site, including the EFSA and UK Food Standards Agency findings. thecattlesite.com
32. US FDA, Center for Veterinary Medicine, letter to Elanco on Bovaer 10 (24 May 2024). aglaw.psu.edu (copy)

Temperature, CO_2 and ONI figures in sections 1, 2 and 5 are computed by this site from NASA GISTEMP v4, NOAA Mauna Loa and NOAA CPC data (see the data archive).

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